



**CALICUT UNIVERSITY – FOUR-YEAR
UNDERGRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	EXPLORATORY DATA ANALYSIS USING PYTHON				
Type of Course	Major Elective (SPECIALIZATION IV: DATA SCIENCE AND ARTIFICIAL INTELLIGENCE)				
Semester	V				
Academic Level	300 -399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Fundamental Programming Concepts in Python 2. Basic idea of Statistics				
Course Summary	This course provides insight into the basic concepts of data analysis and different visualization tools and techniques and teaches the application of these techniques using Python packages.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the types of data and the applications of data science	U	C	Instructor-created exams / Quiz
CO2	Analyse the irregularities present in the data and perform data cleaning	An	C	Problem-solving assessments
CO3	Become familiar with data format & programs used in data analysis	U	F	Practical Assignment / Observation of Practical Skills

CO4	Understand & apply Pandas module for data analysis	U, Ap	P	Instructor-created exams, Practical Assignment / Observation of Practical Skills
CO5	Understand & apply Seaborn module for data visualization	U, Ap	P	Instructor-created exams, Practical Assignment / Observation of Practical Skills
CO6	Learners will develop skills in advanced features of spreadsheets such as macros, protecting data sheets and workbooks, utilizing split, freeze, and hide options effectively	Ap	P	Assignments/ Case Studies
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create(C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Mark 70
I	Introduction to Data Science		8	10
	1	Introduction to Data Science-Definition	2	
	2	Evolution of Data Science	2	
	3	Data Science Roles	2	
	4	Application of data sciences.	2	
	References 1. O'Neil, Cathy, and Rachel Schutt. Doing data science: Straight talk from the frontline. " O'Reilly Media, Inc.", 2013. 2. Machine Learning in Data Science using Python, Dr. R. Nageswara Rao, Dream tech press, 2022 3. Shah, Chirag. A Hands-On Introduction to Data Science. United Kingdom, Cambridge University Press, 2020			

II	Data Collection and Data Pre-Processing		12	18
	5	Data and Data Attributes , Types of Data & Data Attributes	2	
	6	Data Collection Strategies	2	
	7	Data Pre-Processing , Data Cleaning	2	
	8	Data Integration and Transformation	3	
	9	Data Reduction and Discretization	3	
	References			
	Chapter 2, 3 of Book 2			
III	Data Analysis and Manipulation using Pandas		12	18
	10	Introducing different data file formats: csv, xls, tab, dat formats.	2	
	11	Series - constructing from an array, using explicitly defined indices, using a dictionary	2	
	12	Data Frame - constructing from arrays, dictionaries, structured arrays, and series, Indexing of data frames	2	
	13	Arithmetic and Binary operations on Data frame	2	
	14	Broadcasting operations	2	
	15	Universal functions, melt() and pivot()	2	
	References			
	Chapter 5 of Book 1			
IV	Data Visualization using Seaborn		16	24
	16	Review of Data Visualization using matplotlib	2	
	17	Loading datasets in Seaborn, Distribution plot	2	
	18	Count plot, box plot, scatter plot, joint plot.	3	
	19	Line Plot, displaying scatter plot with regression line	2	
	20	Creating subplots	3	
	21	Heat map - cat plot	2	
	22	Violin plot - pair plot.	2	
	References			
	Chapter 6 of Book 1			

V	Open ended Module Hands-on Data Visualization: Working with Pandas data frames Basic plots using Matplotlib Distribution Plots: Histogram, Density Plot, Box Plot, Violin Plot etc Plotting Geospatial Data Introduction to Geoplotlib, The Design Principles of Geoplotlib Geospatial Visualizations - Choropleth Plot, GeoJSON File Introduction to Folium Visualizing Data: Building a Google map from geocoded data Making Things Interactive with Bokeh : Introduction to Bokeh, Concepts of Bokeh, Interfaces in Bokeh Bokeh Server, Presentation, Integrating, Adding Widgets	12	
Books and References 1. Data Science and Machine Learning using Python by Reema Thereja Mc Graw Hill(Book1) 2. Data Mining Concepts and Techniques by Jiawei Han , Elsiever(Book 2) 3. O'Neil, Cathy, and Rachel Schutt. Doing data science: Straight talk from the frontline. " O'Reilly Media, Inc.", 2013. 4. Machine Learning in Data Science using Python, Dr. R. Nageswara Rao, Dream tech press, 2022			

Mapping of COs with PSOs and POs :

	PS O1	PS O2	PSO 3	PS O4	PS O5	PSO 6	PO1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	2	0	1	0	2	0	2	0	1	2	2	0	0
CO 2	1	2	2	1	1	2	2	0	2	2	2	0	0
CO 3	2	1	2	1	1	1	2	0	1	2	2	0	0
CO 4	2	1	2	2	2	2	2	0	2	3	3	0	1

CO 5	2	1	2	2	2	2	2	0	2	3	3	0	1
CO 6	1	2	2	1	1	2	2	0	2	2	2	0	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory /Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	